8-W1　材料非均匀时的稳恒体电荷

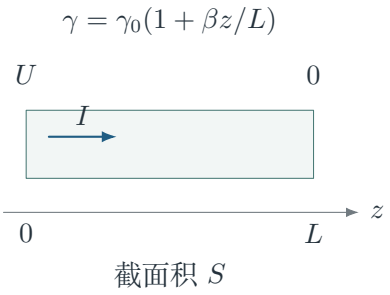


图 8-12　题图　电导率沿导线增大，两端电势固定

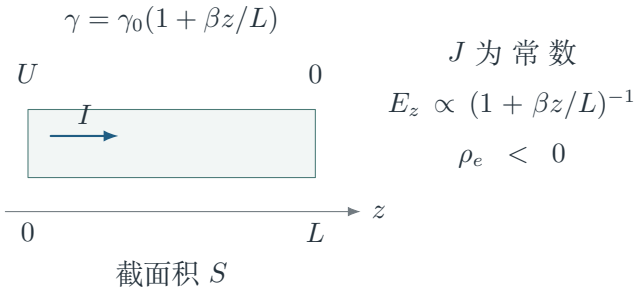
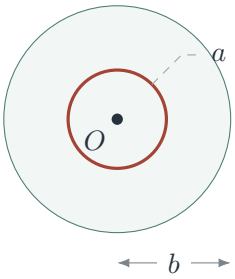


图 8-13　解答分析图　同一电流密度对应递减电场和负体电荷

8-W2　表面电荷梯度与轴向电流

横截面，轴向为 z

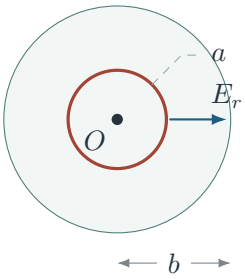


内 侧：$V_0 - E_0 z$

外 侧：0

图 8-14　题图　内导线与外等势筒的同轴边界

横截面，轴向为 z



内 侧：$V_0 - E_0 z$

外 侧：0

$$\lambda \propto V_0 - E_0 z$$

$$I = \gamma \pi a^2 E_0$$

图 8-15　解答分析图　图取 $V_0 - E_0 z > 0$，径向场向外；电势过零后反向

8-W3　由磁针偏角测量电流

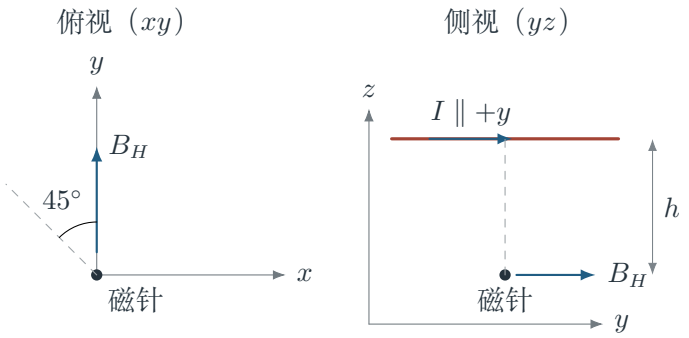


图 8-16　题图　导线电流与背景场平行，侧视图标出针到导线的高度

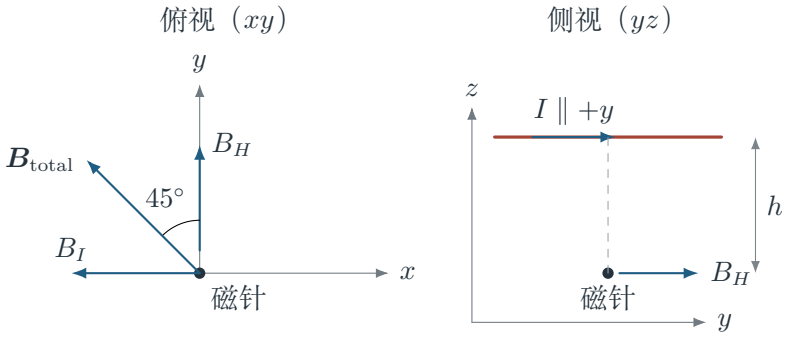


图 8-17　解答分析图　俯视图的导线场沿负 x，磁针沿两个正交场的合场偏转

8-W4　两电流元的形式磁力

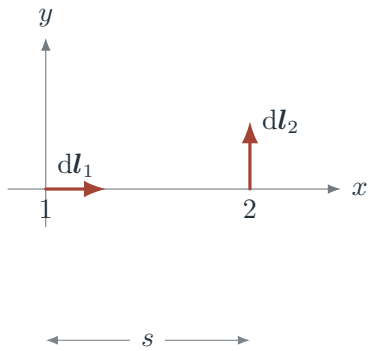


图 8-18　题图　两积分电流元的几何方向

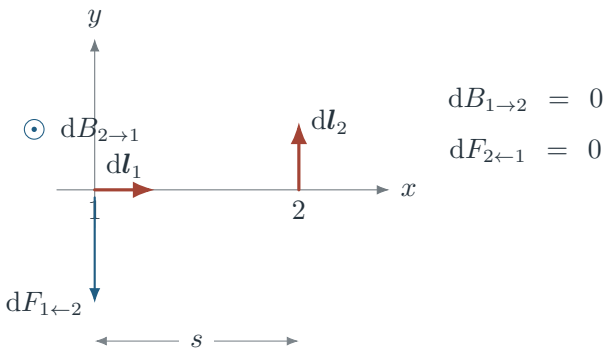


图 8-19　解答分析图　源场与受力各作一次叉乘，元对的形式力不互易

8-W5　不对称有限直导线的磁场

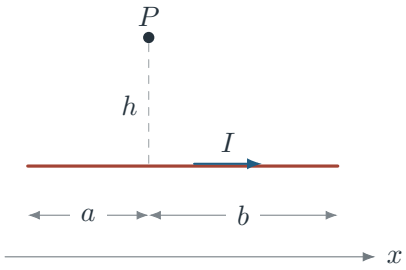
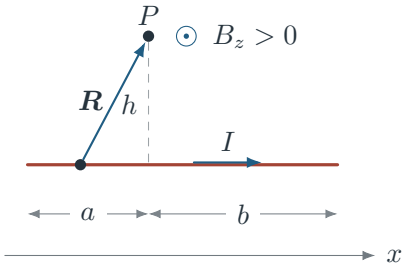


图 8-20　题图　垂足在有限直线段内部，两端长度可以不同



a, b 分别进入端点项

$a, b \to \infty$ ： $B = 2\kappa I/h$

图 8-21　解答分析图　叉乘方向为纸外，端点项保留各自几何长度

8-W6　扇环回路的中心磁场

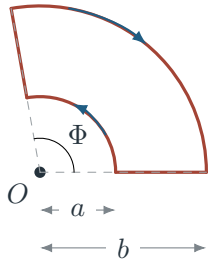
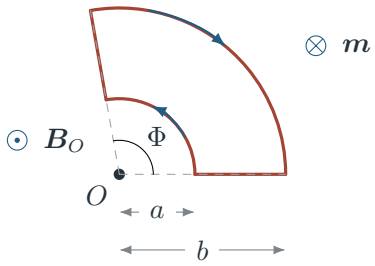


图 8-22　题图　内圆弧逆时针、外圆弧顺时针的闭合扇环



径向连接段：$B_O = 0$

$B_O = \kappa I \Phi (1/a - 1/b)$

图 8-23　解答分析图　中心场沿纸外，回路磁矩沿纸内

8-W7　矩形回路的中心磁场

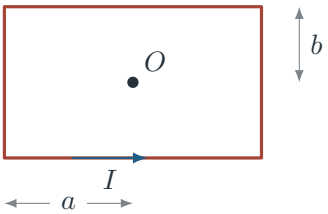
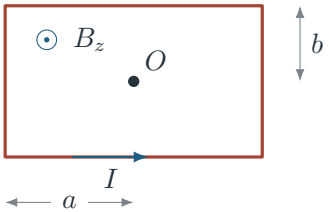


图 8-24　题图　矩形半边长为 a, b，电流逆时针



四边的中心场同向

$$B_O = 4\kappa I\sqrt{a^2+b^2}/(ab)$$

图 8-25　解答分析图　逐边积分，水平边与竖直边的距离不同

8-W8　反向同心线圈的场抵消

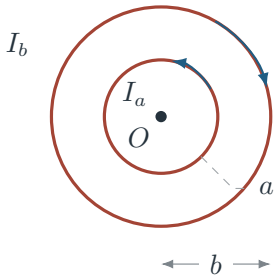


图 8-26　题图　同心反向线圈，外圆半径大于内圆

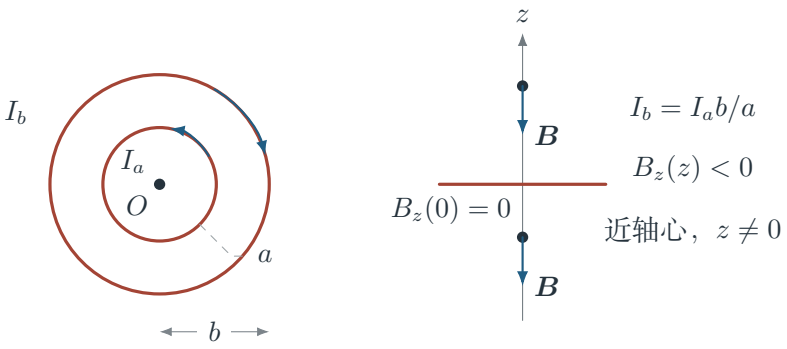


图 8-27　解答分析图　中心场抵消后，轴上仍保留负的二阶变化

8-W9　均匀外场中曲线段的总力

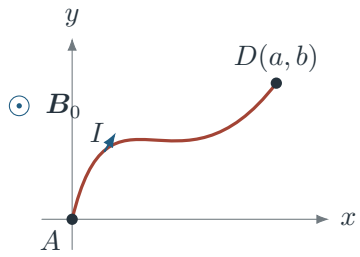
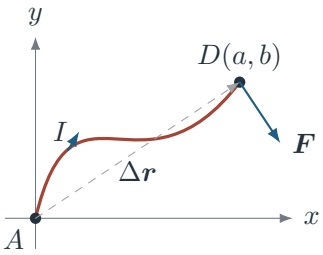


图 8-28　题图　任意曲线段处于均匀纸外磁场



$$F_x = IB_0 b$$
$$F_y = -IB_0 a$$

图 8-29　解答分析图　总力只依赖起终点位移；闭合回路总力为零

8-W10　直导线附近线框的磁力

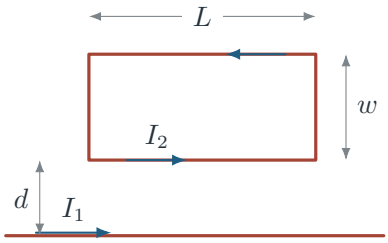


图 8-30　题图　线框近边电流与长直导线同向

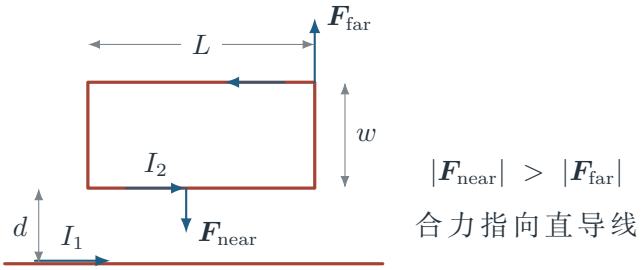


图 8-31　解答分析图　垂直于直导线的两边受力抵消，平行两边给向内合力

8-W11　几何缩放下的磁场与磁力

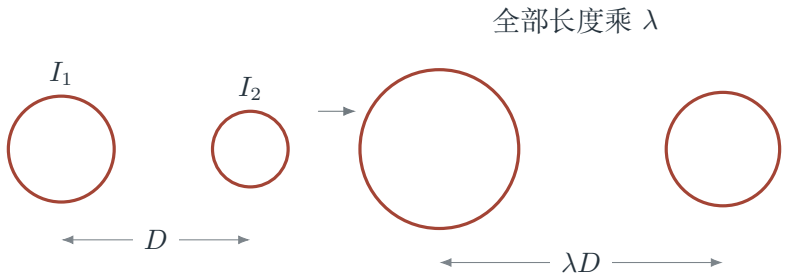
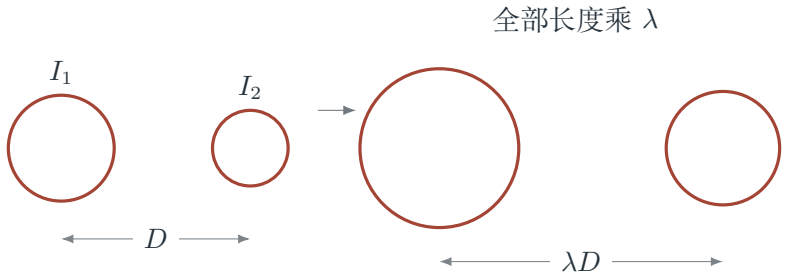


图 8-32　题图　两个完整回路同时等比例缩放，电流保持不变



$$B' = B/\lambda$$

$$F' = F$$

$$\tau' = \lambda\tau$$

图 8-33　解答分析图　力与磁场、受力元长度的缩放共同决定

8-W12　反向双线的远处磁场

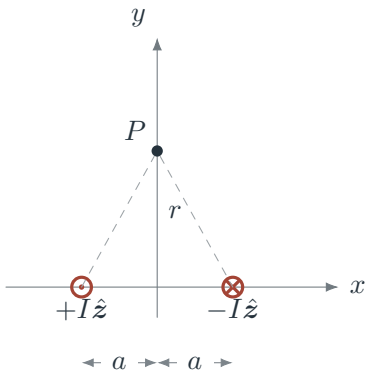
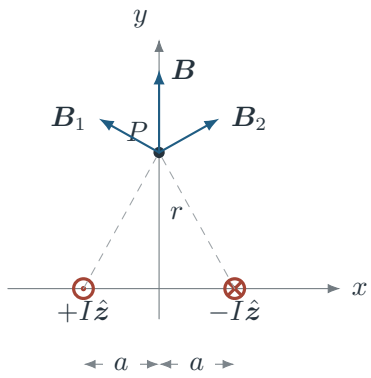


图 8-34　题图　反向平行双线的横截面



$$B_x = 0$$

$$B_y = \mu_0 I a/[\pi(a^2 + r^2)]$$

$$a \ll r \ll \text{端部距离}$$

图 8-35　解答分析图　水平分量相消，竖直分量相加；中间远区按 r^{-2} 衰减

8-T1　电中性与非零电流

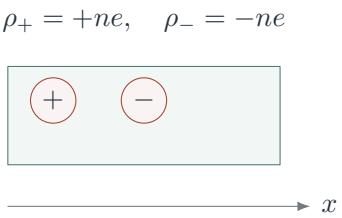


图 8-36　题图　离子静止，电子沿负 x 漂移

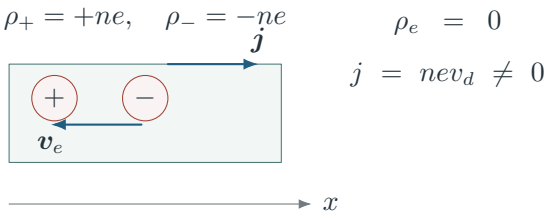


图 8-54　解答分析图　净电荷相消不等于各类电荷的通量相消

8-T2　指数变化的电导率与体电荷

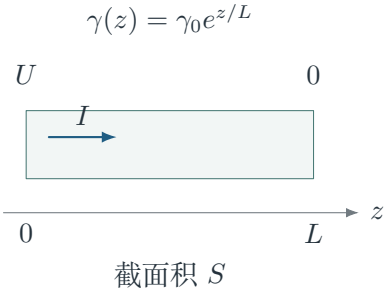


图 8-37　题图　指数电导率下的一维稳恒模型

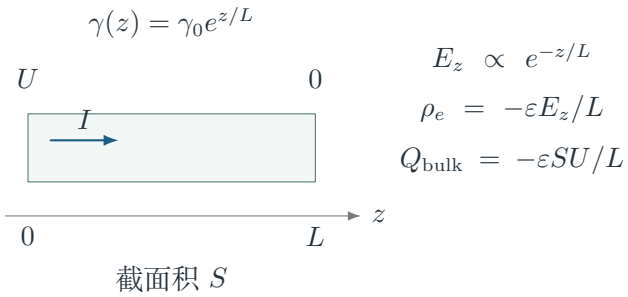


图 8-55　解答分析图　电压积分确定恒定电流密度，场梯度确定体电荷

8-T3　同轴结构中的电势零点

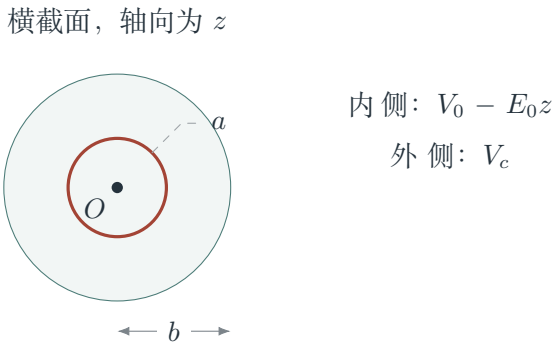


图 8-38　题图　外筒保持常数电势 V_c

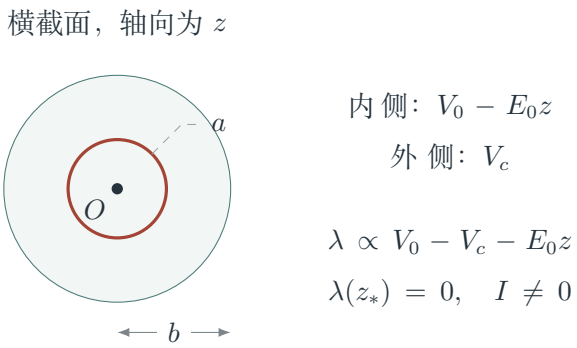


图 8-56　解答分析图　表面电荷由相对电势决定，电流由轴向梯度决定

8-T4　非正交磁场中的磁针偏转

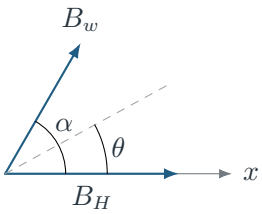


图 8-39　题图　背景场与导线场成锐角，磁针沿合场

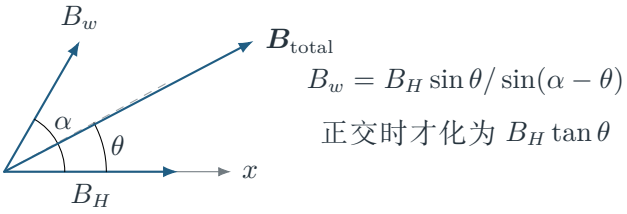


图 8-57　解答分析图　由水平和竖直分量反解导线磁场

8-T5　载流线段的磁力

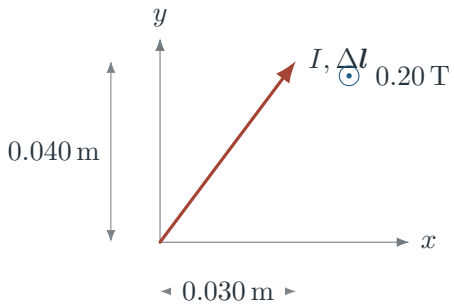
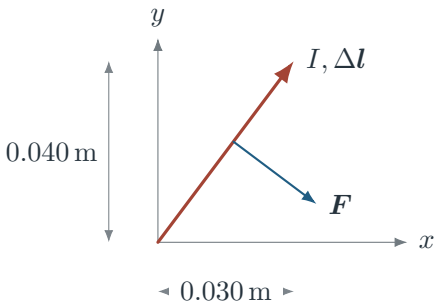


图 8-40　题图　直导线位于 xy 平面，外场垂直纸面



$$\boldsymbol{F} = (0.016, -0.012, 0)\ \mathrm{N}$$

$$\boldsymbol{F} \cdot \Delta\boldsymbol{l} = 0$$

图 8-58　解答分析图　磁力沿右下，与导线端点位移垂直

8-T6　垂足落在线段之外

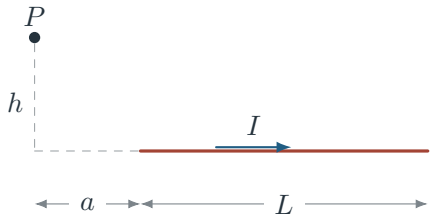
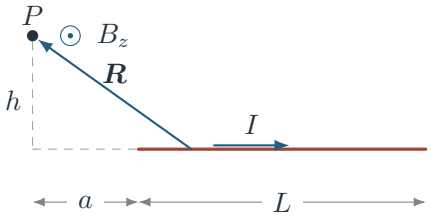


图 8-41　题图　场点的垂足落在有限线段左侧



两端的有向项作差

$L \to 0:\ B \propto L$

图 8-59　解答分析图　两端角不能当成正数直接相加

8-T7　折成直角的长导线

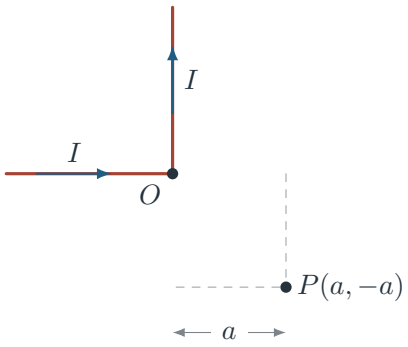
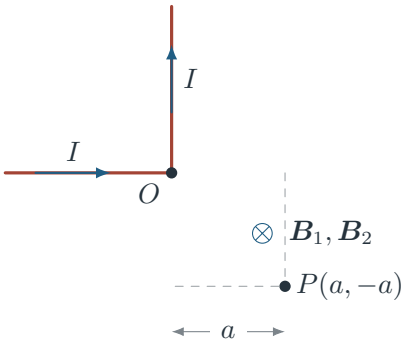


图 8-42　题图　电流从负 x 轴转向正 y 轴



每段：$-\kappa I(1-1/\sqrt{2})/a$

两段场均沿 $-z$

图 8-60　解答分析图　场点在两段电流的右侧，两场均指向纸内

8-T8　大圆弧与弦构成的闭合回路

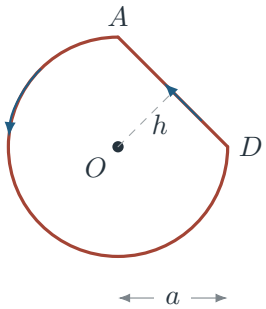
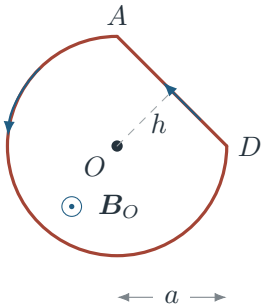


图 8-43　题图　大圆弧逆时针，直弦由右端回到上端



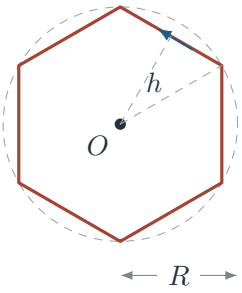
弧 角：$3\pi/2$

弦 场：$2\kappa I/a$

两 场 同 向

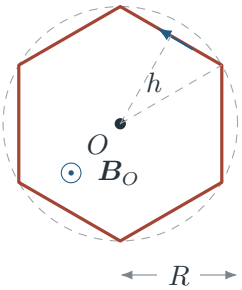
图 8-61　解答分析图　弦电流也绕中心逆时针，其中心场沿纸外

8-T9　正多边形回路的中心磁场



图以六边形示意
题设边数为 n

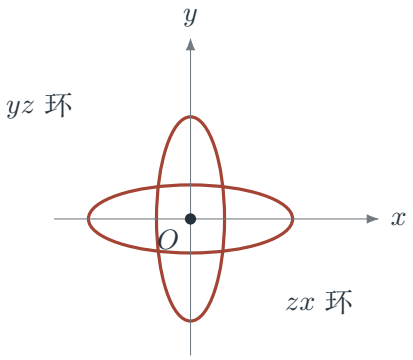
图 8-44　题图　外接圆半径固定的正多边形回路



$$h = R\cos(\pi/n)$$
$$B_O = \mu_0 I n \tan(\pi/n)/(2\pi R)$$

图 8-62　解答分析图　每边的法向场同向，随边数增大趋于圆环场

8-T10　互相垂直的圆线圈



各环法向为 $+x,+y$

半径 R，电流 I

图 8-45　题图　两个互相垂直、中心重合的圆线圈

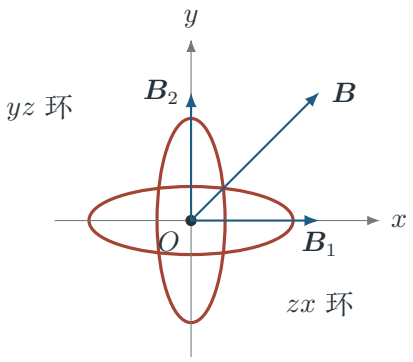


图 8-63　解答分析图　中心场位于两个正法向的角平分线上

8-T11　径向分布的环带电流

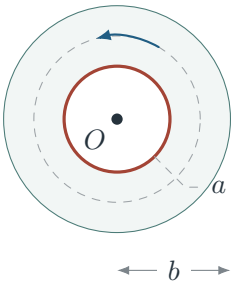
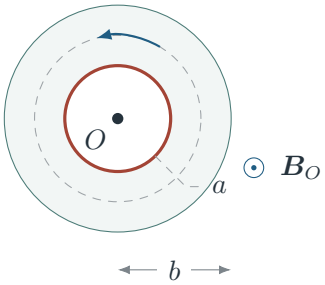


图 8-46　题图　环带总电流均匀分配在径向宽度上



$$\mathrm{d}I = I\,\mathrm{d}r/(b-a)$$

$$\mathrm{d}B = \mu_0 \mathrm{d}I/(2r)$$

图 8-64　解答分析图　逐个细圆环相加，中心场积分产生对数

8-T12　均匀带电圆盘旋转

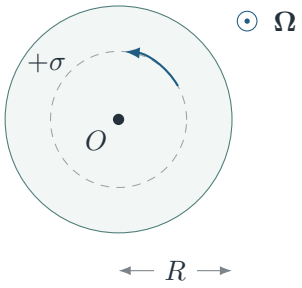
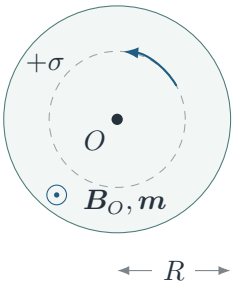


图 8-47　题图　均匀正电荷圆盘绕纸外轴转动

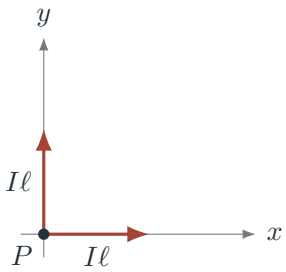


$$\mathrm{d}Q = 2\pi\sigma r\,\mathrm{d}r$$

$$\mathrm{d}I = \sigma\Omega r\,\mathrm{d}r$$

图 8-65　解答分析图　电荷环每转一周通过截面一次，磁场和磁矩均沿纸外

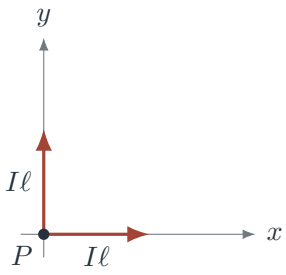
8-T13　两次测力重建磁场



同一点，两次改变测试方向

$I\ell = 0.20\,\mathrm{A\,m}$

图 8-48　题图　两段测试导线分别沿正 x, y



$F_{1y} = -I\ell B_z$

$F_{2x} = I\ell B_z$

$B_x = .010,\ B_y = .020,\ B_z = .030\,\mathrm{T}$

图 8-66　解答分析图　共有分量 B_z 的两次反演可互相核对

8-T14　从完整回路推出磁偶极远场

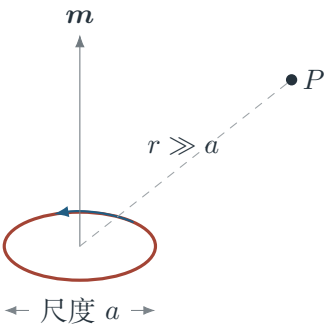
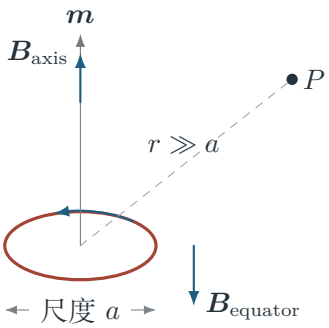


图 8-49　题图　有限局域回路与远处场点



$$\oint \mathrm{d}\boldsymbol{l} \;=\; 0$$

首个一般非零项为偶极场

$$B \;\propto\; r^{-3}$$

图 8-67　解答分析图　轴向场与磁矩同向，赤道场反向；闭合性消去 r^{-2} 项

8-T15　完整回路的作用力与反作用力

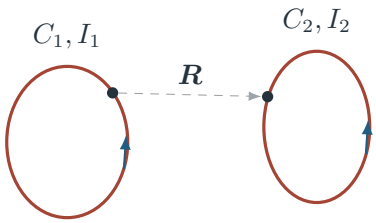
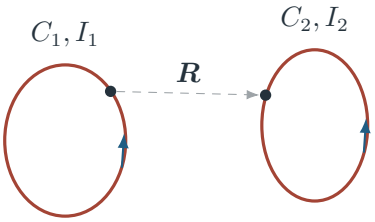


图 8-50　题图　互不相交的两个完整闭合回路



$$\oint_{C_2} \mathrm{d}(1/R) = 0$$

交换编号：$\boldsymbol{R} \to -\boldsymbol{R}$

$$\boldsymbol{F}_1 + \boldsymbol{F}_2 = 0$$

图 8-68　解答分析图　元对的第一项沿闭合回路积分为零，剩余总力互易

8-T16　均匀场中回路的力与力矩

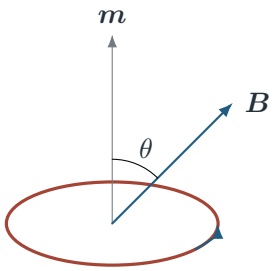
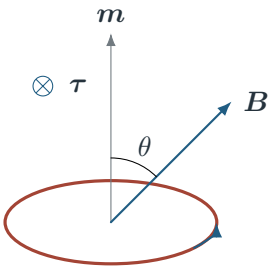


图 8-51　题图　回路磁矩与均匀外场成角



$$\boldsymbol{F}_{\text{total}} = 0$$

$$\boldsymbol{\tau} = \boldsymbol{m} \times \boldsymbol{B}$$

同向朝向稳定

图 8-69　解答分析图　图中力矩沿纸内，推动磁矩转向外场

8-T17　限时综合：亥姆霍兹线圈

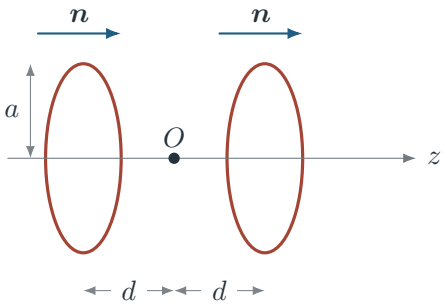
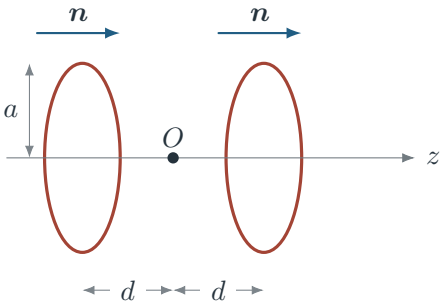


图 8-52　题图　两同向单匝线圈位于对称位置



$$d = a/2$$

$$B_z(0) = (\mu_0 I/a)(4/5)^{3/2}$$

相对变化首项：$-\frac{144}{125}(z/a)^4$

图 8-70　解答分析图　中心间距等于半径时，二阶轴向场变化消失

8-T18　限时综合：磁力吸合的临界条件

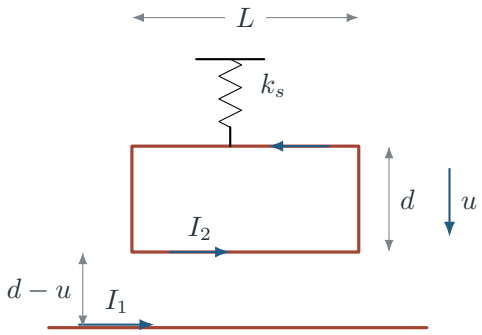


图 8-53　题图　位移向导线为正，线框宽度保持为 d

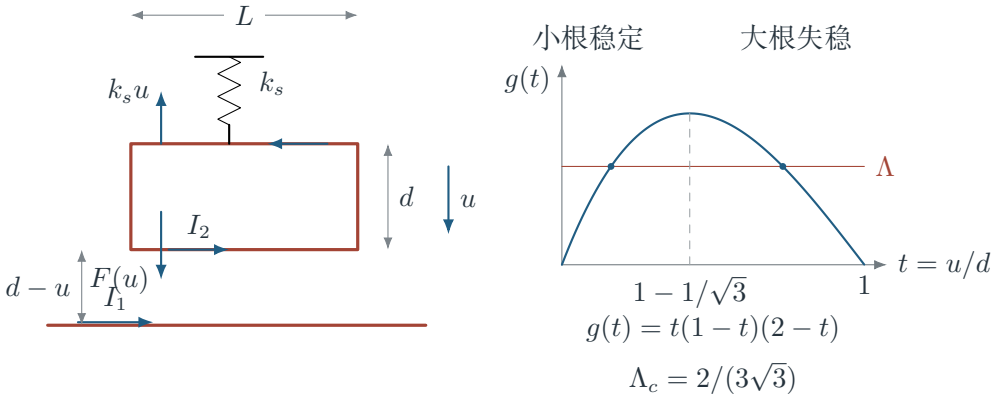


图 8-71　解答分析图　磁力向内、弹簧力向外；两平衡在临界点合并